How Rouge Works: Building Epoch from Scratch

Rouge is an autonomous product development system. It takes a product idea, refines it through interactive Slack conversation, then runs an iterative build-evaluate-improve loop until the product meets a quality bar. No human writes code. No human triages bugs. The system builds, tests, evaluates, and ships — pausing only when it needs a product decision it cannot make itself.

Let's watch it build a product from scratch.

The Stack

Rouge is a Node.js orchestrator that drives Claude via the Anthropic API to build, test, and evaluate web products. Every product it ships comes with a full production stack, provisioned automatically.

How it runs

A Node.js launcher (`rouge-loop.js`) orchestrates the Karpathy Loop. For each phase, it calls the Claude API with a phase-specific system prompt, the project context, and the working directory. Claude reads the codebase, writes code, runs tests, and commits — all through API tool use. The launcher monitors progress, handles rate limits, manages state transitions, and advances to the next phase.

The Slack bot (`bot.js` , Bolt.js in Socket Mode) provides the human control plane: seeding new products, monitoring progress, pausing/resuming builds, approving final reviews, and sending feedback.

What gets deployed

Every Rouge-built product deploys to this stack:

Layer	Tool	Provisioned by
Hosting	Cloudflare Workers (via OpenNext)	Rouge — automatic
Database	Supabase (Postgres + Auth + Realtime)	Rouge — automatic (slot rotation for free tier)
Payments	Stripe	Rouge — sandbox auth, human activates production
Error monitoring	Sentry	Rouge — auto-creates project, writes DSN to env
Product analytics	PostHog (EU hosting)	Rouge — shared project, product-tagged events
Session recording	PostHog	Included with analytics
CI/CD	GitHub Actions (lint, type check, test, build)	Template — ships with project
Security headers	Next.js middleware (CSP, HSTS, X-Frame-Options)	Template — ships with project
i18n	next-intl (key-based, Claude-translated)	Template — ships with project
Legal pages	Privacy policy + Terms of Service scaffolds	Template — ships with project
Browser QA	GStack (headless browser for automated testing)	Rouge — used during product-walk phase

What Rouge manages automatically

- **Deployment**: `npm run build` → `opennext build` → `wrangler deploy` with post-deploy health check and automatic rollback on failure
- **Database migrations**: dry-run preview, destructive operation detection (DROP/TRUNCATE blocked), audit trail
- **State snapshots**: before each phase, state.json and cycle_context.json are snapshotted for corruption recovery
- **Cost tracking**: self-calibrating cost estimation from actual phase timings
- **Cross-product learning**: after each completed product, learnings feed into a personal Library that calibrates future builds

Slack integration

Rouge is controlled via Slack. The bot runs in Socket Mode (no public URL needed) and provides:

- **Seeding**: `@Rouge` in a channel starts a seeding conversation. The 8-persona swarm runs through Slack threading — each discipline posts in the thread, and the human responds in the same thread.
- **Monitoring**: The App Home dashboard shows every project with its current state, cycle number, health score, confidence level, staging URL, and feature area progress. Buttons for Start, Pause, Resume, Skip, and Ship It.
- **Notifications**: Phase transitions, QA results, screenshots, and escalations post as rich Block Kit messages with context (not bare text). Rate limit backoffs and timeouts are logged.
- **Control**: `/rouge start`, `/rouge pause`, `/rouge resume`, `/rouge ship`, `/rouge feedback` — all from Slack. No SSH, no terminal, no deployment dashboard.
- **Final review**: When a product reaches final-review, the human gets a Slack notification with the staging URL. They test it at their own pace and either `/rouge ship` to approve or `/rouge feedback` to request changes.

The launcher and Slack bot are separate processes. The launcher writes state; the bot reads it and sends notifications. Human commands (pause, resume, ship) write to state.json, and the launcher picks up the change on the next iteration.

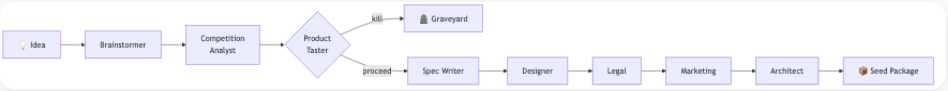
The Idea

The product: **Epoch** — a focus timer that's beautiful enough to leave on screen. A Pomodoro timer for knowledge workers who have tried every timer app and found them either ugly or bloated.

The idea entered Rouge through **Rouge Spec**, the interactive seeding phase. This is where a human and AI co-design a product through Slack conversation — not a requirements document, but a dialogue.

How Rouge Spec Works

Rouge Spec runs a **seeding swarm**: eight discipline-specific AI personas that each interrogate the idea from a different angle. They don't run simultaneously — they take turns, building on each other's output:



The **human participates throughout** — answering questions, making taste decisions, vetoing bad ideas at any stage. The swarm asks; the human decides. The Product Taster is the hard gate: ideas that aren't worth building go to the graveyard before any specs are written.

The eight personas in order:

#	Persona	Role
1	**Brainstormer**	Expand the idea — 10-star version, emotional core, target user
2	**Competition Analyst**	Map the landscape — what exists, where's the gap
3	**Product Taster**	Challenge the idea — kill it or scope it

4	**Spec Writer**	Convert vision into feature areas + acceptance criteria
5	**Designer**	Three-pass: UX architecture → components → visual design
6	**Legal Advisor**	Privacy, terms, compliance
7	**Marketing Strategist**	Positioning, landing page, launch strategy
8	**Technical Architect**	Stack selection, infrastructure, deployment target

When the conversation converges, the seed package contains:

- **Vision document** — emotional north star, aesthetic direction, anti-goals
- **Feature area specifications** — with acceptance criteria per area
- **Design artifact** — style tokens, screen mappings, interaction specs
- **Infrastructure plan** — deployment target, database needs, auth requirements

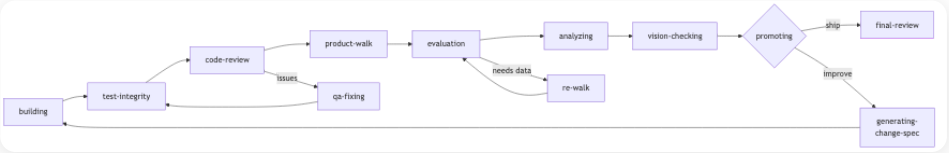
For Epoch, this conversation established:

- **Emotional north star**: "From 'where did the last 3 hours go?' to 'I can see my focus happening.'"
- **Aesthetic**: Techno-futuristic. Frosted glass, subtle glows, precise typography. Linear's design language meets a high-end watch face.
- **Scope**: Full Pomodoro cycle (focus, short break, long break), configurable settings, session counter, keyboard shortcuts, synthesized audio chime. No accounts, no history beyond today, no themes, no task lists.
- **Differentiator**: Optimizes for *presence* — how it feels on your screen — not for features. The design IS the product.

The seed package contained six feature areas with 37 acceptance criteria, a complete design system (color palettes per phase, typography scale, glass effects), and deployment target (Cloudflare Workers, static Next.js export). The system estimated one Rouge cycle for the initial build. It took five.

The Karpathy Loop

Rouge builds products through an iterative loop inspired by Andrej Karpathy's description of how neural networks train: build, evaluate, adjust, repeat. Each cycle advances the product toward convergence with the vision.



Each phase has a distinct role:

Phase	Purpose	Uses Browser?
building	Write code, run tests, deploy to staging	No
test-integrity	Verify all tests pass, no regressions	No
code-review	Engineering lens — architecture, dead code, security	No
product-walk	Pure observation — screenshot every state, record what exists	Yes
evaluation	Three judges score the walk data (QA, Design, PO lenses)	No
analyzing	Decide: ship, improve, or escalate	No
vision-checking	Compare built product against original vision	No
promoting	Advance to next cycle or trigger final review	No

final-review	Holistic end-of-project walkthrough	Yes

The critical design choice: the browser opens exactly once per cycle (product-walk), and all three evaluation lenses judge the same observation data. This is the **observe-once, judge-through-lenses** architecture.

Cycle 1: The First Build

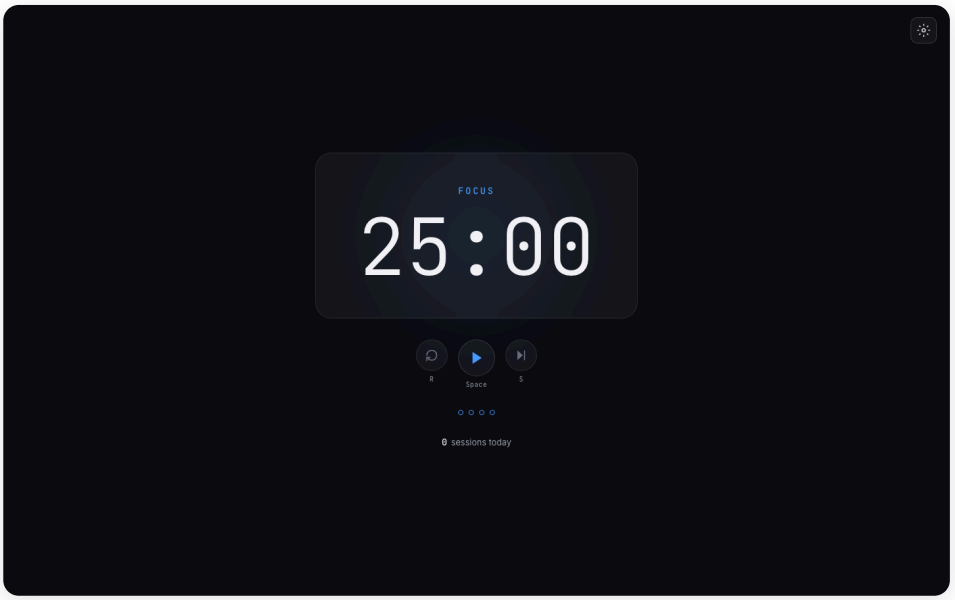
Rouge built all six feature areas in a single cycle:

1. **Pomodoro engine** — State machine for focus/short-break/long-break cycling, localStorage persistence, Web Audio API chime synthesis
2. **Timer display** — Responsive monospace digits (72-120px via CSS clamp), phase-colored frosted glass card
3. **Timer controls** — Play/pause, reset, skip with keyboard shortcuts (Space, R, S)
4. **Cycle indicator** — Four dots showing position in the Pomodoro cycle
5. **Session counter** — Daily focus session count, persisted in localStorage
6. **Settings modal** — Eight configurable parameters (durations, auto-start, sound, notifications)

The builder made architectural decisions along the way:

- **Web Audio API** over audio files for the chime — zero dependencies, matches the futuristic aesthetic
- **CSS Modules** over Tailwind — natural syntax for complex effects (backdrop-filter, radial gradients, keyframe animations)
- **Date.now()-based timing** over setInterval — immune to background tab throttling
- **Static export** — the entire app is client-rendered, no SSR needed

Result: 89 tests passing. Deployed to Cloudflare Workers. The product loaded, the timer counted down, the phase transitions shifted the entire color palette.



But the quality evaluation found gaps.

Cycle 2: The Quality Ratchet

The product-walk and evaluation phases identified concrete issues:

- **Missing error boundary** — no crash recovery UI
- **Settings modal used a div overlay** — no focus trapping, no Escape handling, screen readers couldn't distinguish it from the page
- **Footer text contrast**: 3.2:1 ratio against dark background, failing WCAG AA (needs 4.5:1)
- **No semantic landmarks** — screen readers had no navigation structure
- **Keyboard shortcuts undiscoverable** — no visual hints for R/Space/S

The system generated change specs for each issue and the builder implemented fixes:

- ErrorBoundary with styled dark fallback UI

- Native `<dialog>` element replacing the div overlay (automatic focus trap, background inertness, Escape)
- Semantic HTML5 landmarks (`<header>` , `<main>` , `<footer>`)
- Inline keyboard shortcut hints below control buttons
- Contrast fixes (session counter text from #6b7280 to #9ca3af)

After cycle 2, the health score jumped from 68 to 82.

Lighthouse progression across cycles

Metric	Cycle 1	Cycle 5
Performance	91	91
Accessibility	89	**100**
Best Practices	96	**100**
SEO	100	100
Tests passing	89	90
Spec criteria pass rate	—	100% (67/67)

What the System Sees

Rouge does not evaluate products through vibes. Every judgment is structured data with evidence. Here is what each lens produces.

Spec criterion check

Each acceptance criterion is evaluated with a pass/fail status and the specific evidence that supports the verdict:

```
{
  "id": "AC-display-1",
```

```
  "criterion": "Timer displays MM:SS in monospace font >=
  "status": "pass",
  "evidence": "Timer uses clamp(72px, 15vw, 120px) — minim
}
```

67 of these are checked every cycle. Zero are skipped.

QA health score breakdown

The health score is a weighted composite. Points are deducted for specific findings at three severity levels:

Category	Score	Detail
Criteria compliance	100%	67/67 pass
Functional correctness	Clean	0 console errors, 803ms load, all controls verified
Code quality (AI audit)	90/100	Architecture 88, Consistency 92, Robustness 84, Security 96
HIGH findings	0	None remaining (2 resolved in cycle 2)
MEDIUM findings	4	-8 points (e.g., useTimer god hook at 157 lines)
LOW findings	10	-10 points (capped)
Final health score	**82**	

PO review — journey quality

The Product Owner lens evaluates user journeys on four dimensions:

Journey	Clarity	Feedback	Efficiency	Delight	Avg	Verdict

First-time landing	9	8	10	8	9.0	polished
Start/pause focus	9	9	10	8	8.83	polished
Phase transition	9	9	9	9	8.88	polished
Reset current phase	7	9	10	6	8.0	polished
Configure settings	—	—	—	—	8.44	polished
Track daily progress	—	—	—	—	8.5	polished

Overall PO confidence: **0.89**. Verdict: **PRODUCTION_READY**.

QA Tooling

Rouge doesn't just ask Claude "does this look right?" — it runs real tools against real deployed code and feeds structured results into the evaluation.

Browser testing — GStack

GStack is a headless Chromium browser controlled via a single-binary CLI. During the product-walk phase, Claude drives it to navigate every screen, click every element, fill forms, test keyboard navigation, and capture screenshots. Commands are simple:

```
$B goto https://countdowntimer-staging.gregj64.workers.dev
$B click button:has-text("Start")
```

```
$B screenshot screenshots/cycle-3/timer-running.png
$B eval document.querySelectorAll('[aria-label]').length
```

This is how Rouge verifies that buttons actually work, modals actually open, and forms actually validate — not by reading code, but by using the product.

Performance — Lighthouse

GStack runs Lighthouse audits on key pages, producing scores for Performance, Accessibility, Best Practices, and SEO. These scores are tracked across cycles — a regression triggers a finding. Epoch's progression:

Metric	Cycle 1	Cycle 5
Performance	91	91
Accessibility	89	**100**
Best Practices	96	**100**
SEO	100	100

Static analysis — code-review phase

The code-review phase runs CLI tools directly, without a browser:

Tool	What it checks
ESLint	Lint errors and warnings
jscpd	Code duplication percentage
madge	Circular dependency detection
knip	Dead code and unused exports
npm audit	Known dependency vulnerabilities

AI code audit	Seven-dimension review: architecture, consistency, robustness, production risks, security, dead/hallucinated code, tech debt

Each tool produces structured output. The evaluation phase reads all of it and computes the health score.

Accessibility

Accessibility is tested through three channels:

1. **Lighthouse a11y score** — automated WCAG checks
2. **GStack keyboard testing** — Tab through every element, verify Enter/Space activation, check focus trapping in modals
3. **Accessibility tree inspection** — `$B eval` captures landmarks, heading hierarchy, ARIA labels, and roles

This is how Rouge found the missing focus trap in Epoch's settings modal — it literally tabbed through the UI 11 times and observed the focus escaping to the background.

Visual and copy quality

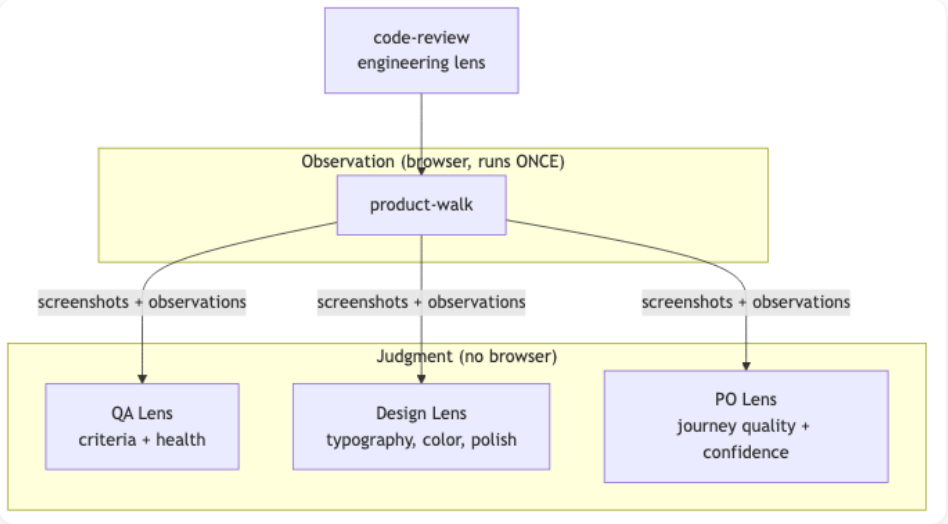
The evaluation phase's Design lens scores screenshots across eight categories (typography, color, spacing, layout, components, interaction, content, polish) and runs an AI slop detector to flag generic or AI-generated-feeling design. The Content & Copy sub-check evaluates whether UI text sounds human, is tonally consistent, and matches the product's personality.

The Architecture: Observe Once, Judge Through Lenses

The original architecture ran QA and PO review as separate phases, each launching its own browser session. This was wasteful — the browser is slow

and expensive, and both phases looked at the same product. Worse, observations could diverge between sessions if state differed.

The redesigned architecture separates observation from judgment:



code-review runs first, using only CLI tools (linting, dependency audit, AI code analysis). No browser. It produces an engineering assessment: architecture score, dead code count, security findings.

product-walk opens the browser once, navigates every screen state (focus, short break, long break, settings open, settings closed, mobile viewport, desktop viewport), takes screenshots, and records structured observations. Then the browser closes.

evaluation runs three lenses on the walk data — QA, Design, and PO — without touching a browser. Each lens scores independently. If any lens needs more information, a targeted **re-walk** opens the browser for specific screens only.

This architecture cut browser time per cycle from ~26 minutes to ~8 minutes and eliminated observation divergence between phases.

What We Learned

The countdowntimer run was Rouge's first end-to-end test. It exposed several system-level problems that were fixed during and after the run.

Rate limits and wasted executions

Rouge calls the Claude API for each phase. During cycle 2, the launcher hit API rate limits and responded by retrying immediately — 95 times. Each retry was a wasted execution that burned tokens and produced no useful output.

Fix: Smart backoff. The launcher now detects rate limit responses in stdout, parses the reset timestamp, and sleeps until the limit clears. A single `sleep` call replaces 95 wasted retries.

Hard timeouts vs. progress-based watchdog

The original launcher used fixed 30-minute timeouts per phase. Some phases (building) legitimately need more time. Others (analyzing) should finish in 2 minutes.

Fix: Progress-based watchdog. The launcher monitors output — if a phase produces no new output for N minutes, it's considered stalled. Active phases that are making progress are never killed prematurely.

Feature-area cycling was wasteful

In cycles 2-4, the builder was instructed to "build all feature areas." For areas already completed in cycle 1, this produced no-op builds — the system would examine the code, confirm it was already done, and move on. Each no-op still cost time and tokens.

Fix: No-op detection. The system tracks which feature areas are complete. When a builder cycle produces no changes for an area, it is marked complete and skipped in future cycles. The builder focuses only on change specs and unfinished areas.

Monolithic QA

The original QA phase tried to do everything: run tests, open a browser, check criteria, score health, evaluate design, assess journeys. It was a 40-minute monolith that frequently timed out or produced partial results.

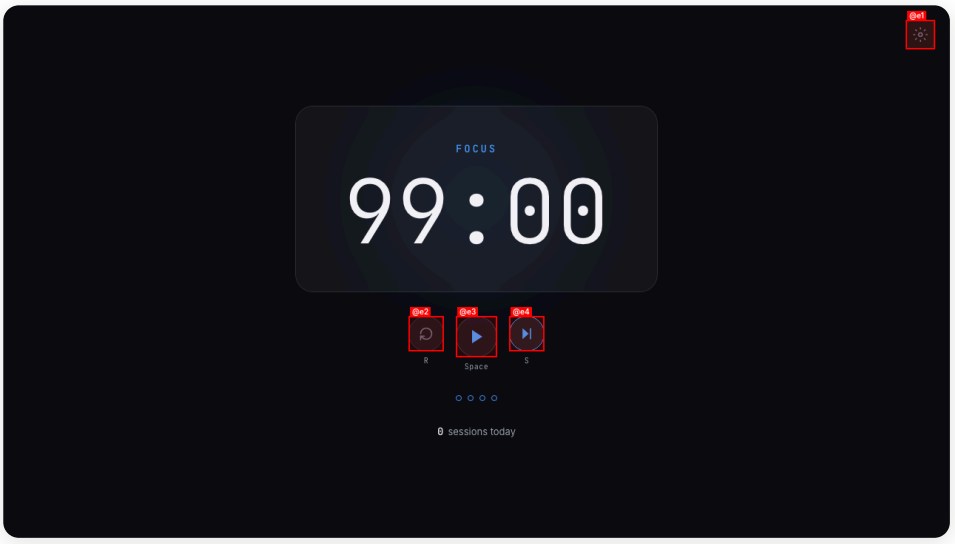
Fix: Split into focused phases (code-review, product-walk, evaluation). Each phase does one thing well. The total time decreased and the quality of assessment increased because each phase has a clear, bounded scope.

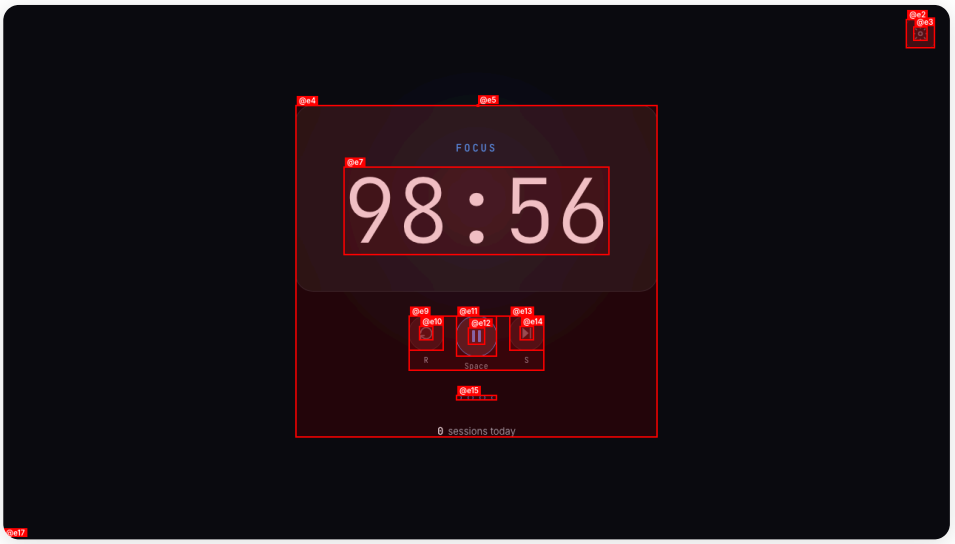
Efficiency

Before fixes: **59% efficiency** — 41% of execution time was wasted on retries, no-ops, and timed-out monolithic phases. After the observe-once architecture and smart backoff: significantly improved, with most waste eliminated at the system level.

The Final Product

After 5 cycles, Rouge produced a deployed Pomodoro timer that met its vision: atmospheric, opinionated, crafted.

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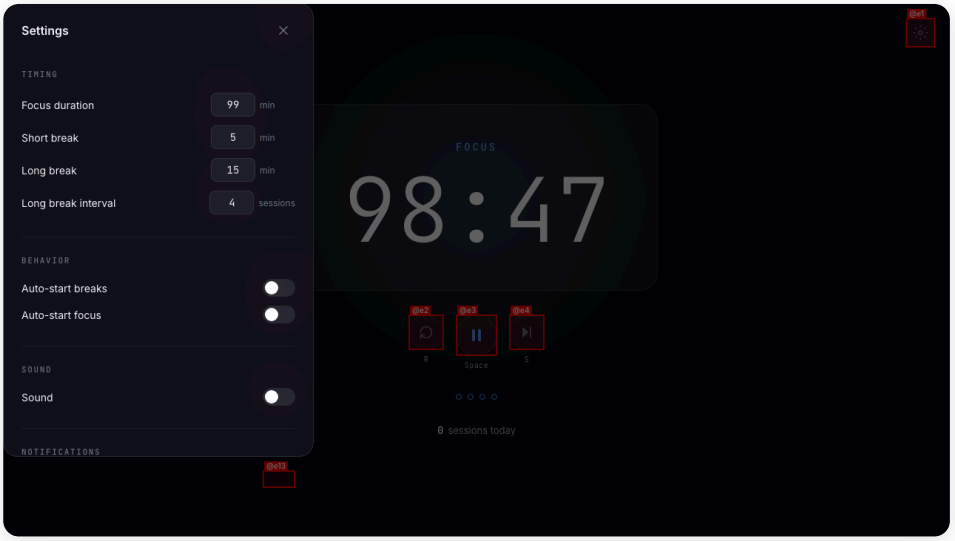
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TIMING
Focus duration
99
min
Short break
5
min
Long break
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min
Long break interval
4
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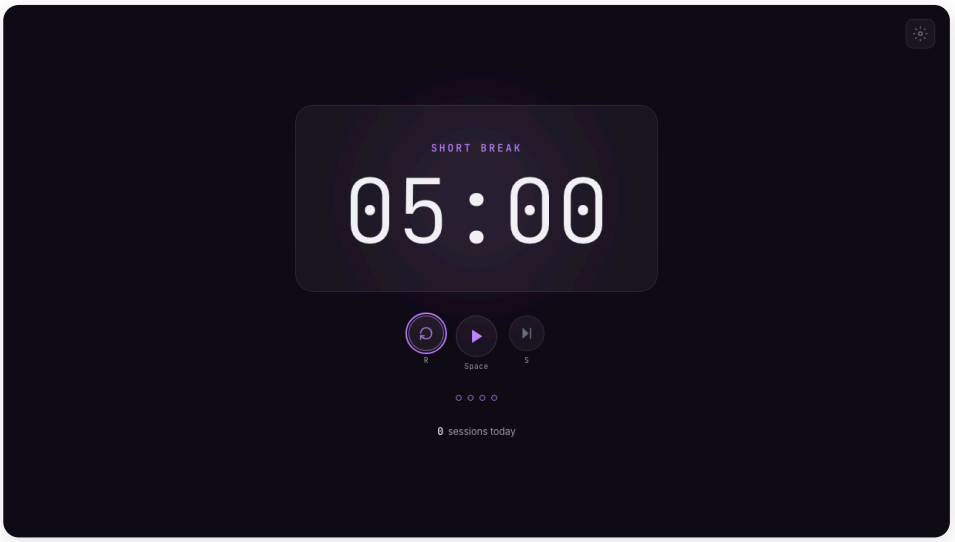


Vision alignment score: **0.91** (converging). The vision check summary:

> "Epoch has converged strongly on its vision. The core promise — a focus timer beautiful enough to leave on screen — is delivered through a cohesive techno-futuristic aesthetic, atmospheric phase transitions, and purposeful minimalism. The product's trajectory is clear: it has become what the vision described."

Key final numbers:

- 90 tests passing
- 67/67 acceptance criteria at 100%
- Health score: 82/100
- Lighthouse: 91 / 100 / 100 / 100
- AI code audit: 90/100
- Design score: 86/100 (AI slop score: 6/100 — the product does not feel AI-generated)
- PO confidence: 0.89
- All 6 user journeys rated "polished"

Cost

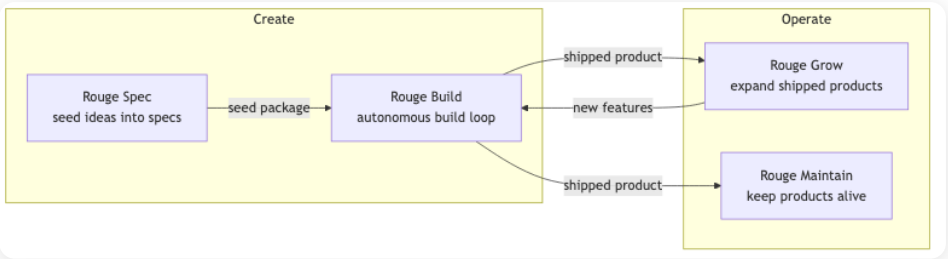
The countdowntimer run consumed approximately **$8.53 of Opus compute time** across 108 minutes of active execution (5 cycles, multiple phases each). This includes building, testing, deploying, evaluating, generating change specs, and re-building.

For context: a human developer building this product (design, implementation, testing, accessibility hardening, deployment) would likely spend 1-2 weeks. At typical contractor rates ($100-150/hour), that's $4,000-12,000. Rouge produced a comparable result for under $10 in compute — three orders of magnitude cheaper.

The real cost was in system development: building Rouge itself. But that cost amortizes across every product Rouge builds after the first.

The Rouge Product Line

The countdowntimer run validated Rouge's core loop. But building a product from scratch is only one capability. The full vision is a product line covering the entire lifecycle:



Rouge Spec — "What to build"

Interactive seeding via Slack. Human + AI co-design a product through conversation with eight discipline-specific personas. Produces a vision, specs, design artifacts, and infrastructure plan.

Rouge Build — "How to build it"

The Karpathy Loop. Takes a seed package and autonomously builds the product through iterative cycles of building, testing, evaluating, and refining. This is what built Epoch.

Rouge Grow — "Make it better"

Feature expansion on shipped products. Unlike Build (which creates from zero), Grow works with existing users, existing data, and existing patterns that must be preserved. It reads analytics (PostHog), user feedback, and market signals to decide what to build next. Then it runs a modified loop that respects backwards compatibility and existing user expectations.

Rouge Maintain — "Keep it alive"

Autonomous production upkeep. SBOM and CVE scanning, bug triage from Sentry error streams, dependency updates, database migration management, SSL certificate renewal, performance regression detection. No new features — just keeping the lights on and the foundation solid.

What We're Building Toward

The countdowntimer was a 6-screen timer app. The architecture now supports much larger projects:

- **Module hierarchy**: Large products decompose into modules with dependency-aware build ordering. A billing module that depends on auth waits until auth is complete. Rouge has a DAG-based resolver for this.
- **Cross-product learning**: The personal Library layer accumulates intelligence across projects. Standards discovered building Epoch (e.g., "use native `<dialog>` for modals", "Date.now() for timers") become calibration data for future products. Product #11 benefits from everything learned building products #1 through #10.

- **Cost predictability**: Before a build starts, the cost estimation engine projects low/mid/high bounds based on project complexity and historical calibration data. "This SaaS will cost $50-150 in compute" — then the human decides whether to proceed.
- **Production readiness from day one**: Every Rouge-built product ships with analytics (PostHog), error monitoring (Sentry), security headers, CI/CD, legal pages, and i18n support. Not bolted on after — baked into the scaffold.

The goal: a factory where you describe a product over coffee, approve a cost estimate, and come back to a deployed, monitored, production-ready application. Rouge handles everything between the idea and the first user.